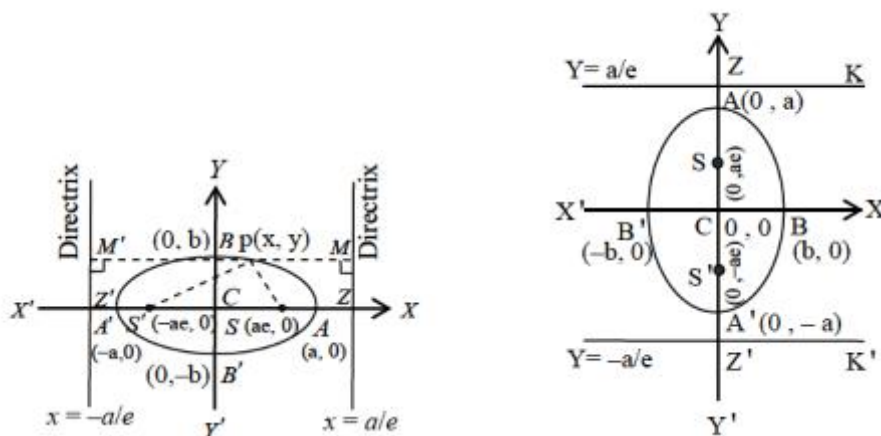


ELLIPSE

EQUATION	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ ($a > b$)	$\frac{(x-\sim)^2}{a^2} + \frac{(y-\)^2}{b^2} = 1$ ($a > b$)	$\frac{(x-\sim)^2}{b^2} + \frac{(y-\)^2}{a^2} = 1$ ($a > b$)
Centre (C)	(0,0)	(0,0)	($\sim$, $\)$	($\sim$, $\)$
Vertices (A, A ¹)	($\pm a, 0$)	(0, $\pm a$)	($\sim \pm a$, $\)$	($\sim$, $\pm a$)
Foci (S, S ¹)	($\pm ae, 0$)	(0, $\pm ae$)	($\sim \pm ae$, $\)$	($\sim$, $\pm ae$)
Point of intersection of Axis and directrix (Z, Z ¹)	($\pm a/e, 0$)	(0, $\pm a/e$)	($\sim \pm a/e$, $\)$	($\sim$, $\pm a/e$)
End point of Latus rectum	($\pm ae, \pm b^2/a$)	($\pm b^2/a, \pm ae$)	($\sim \pm ae$, $\pm b^2/a$)	($\sim \pm b^2/a$, $\pm ae$)
Equation of major axis	$y = 0$	$x = 0$	$y =$	$x = \sim$
Equation of minor axis	$x = 0$	$y = 0$	$x = \sim$	$y =$
Equation of Latera recta	$x = \pm ae$	$y = \pm ae$	$x = \sim \pm ae$	$y = \pm ae$
Equation of directrices	$x = \pm a/e$	$y = \pm a/e$	$x = \sim \pm a/e$	$y = \pm a/e$
Length of major axis	$2a$	$2a$	$2a$	$2a$
Length of minor axis	$2b$	$2b$	$2b$	$2b$
Length of latus rectum	$2b^2/a$	$2b^2/a$	$2b^2/a$	$2b^2/a$
Eccentricity (e)	$\sqrt{\frac{a^2 - b^2}{a^2}}$	$\sqrt{\frac{a^2 - b^2}{a^2}}$	$\sqrt{\frac{a^2 - b^2}{a^2}}$	$\sqrt{\frac{a^2 - b^2}{a^2}}$
Sum of focal distances of a point P on the ellipse	$2a$	$2a$	$2a$	$2a$
Distance between foci	$2ae$	$2ae$	$2ae$	$2ae$
Distance between Vertices	$2a$	$2a$	$2a$	$2a$
Distance between Directrix	$2a/e$	$2a/e$	$2a/e$	$2a/e$



- **Parametric Form of Ellipse:** $x = a \cos \theta$, $y = b \sin \theta$ where $0 \leq \theta \leq 2\pi$.
- **Position of a point with respect to an Ellipse:** Let $P(x_1, y_1)$ be a point and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be an ellipse, then
 - (a) $P(x_1, y_1)$ lies on ellipse if $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 = 0$.
 - (b) $P(x_1, y_1)$ lies inside ellipse if $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 < 0$.
 - (c) $P(x_1, y_1)$ lies outside ellipse if $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 > 0$.
 - (d) Distance of point $P(x_1, y_1)$ from the focus $F_1 = a - ex_1$.
 - (e) Distance of point $P(x_1, y_1)$ from the focus $F_2 = a + ex_1$.
 - (f) $P(x_1, y_1) = (a \cos \theta, b \sin \theta)$ is point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) then $SP = |a - ex_1| = |a - ae \cos \theta|$ and $S^1P = |a + ex_1| = |a + ae \cos \theta|$ where S and S^1 are the foci of the ellipse.
- **Intersection of a line and an Ellipse:** The line $y = mx + c$ intersects the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 - (1) At two distinct points if $a^2m^2 + b^2 > c^2$.
 - (2) At one point if $a^2m^2 + b^2 = c^2$.
 - (3) Does not intersect if $a^2m^2 + b^2 < c^2$.

• **Equation of Tangents to Ellipse:**

- (1) From an external point we can draw only two tangents to an ellipse.
- (2) **Point Form:** The equation of the tangent at $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = 0$.
- (3) **Slope Form:** If the line $y = mx + c$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $a^2m^2 + b^2 = c^2$. Hence the straight line $y = mx \pm \sqrt{a^2m^2 + b^2}$ always represents the tangents to the ellipse.
- (4) **Parametric Form:** The equation of tangent at any point $(a \cos^-, b \sin^-)$ is $\frac{x}{a} \cos^- + \frac{y}{b} \sin^- = 1$.
- (5) The condition of the line $y = mx + c$ may be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 + b^2$ and the point of contact is $\left(\frac{-a^2m}{c}, \frac{b^2}{c} \right)$.
- (6) The condition of the line $lx + my + n = 0$ may be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $a^2l^2 + b^2m^2 = n^2$ and the point of contact is $\left(\frac{-a^2l}{n}, \frac{-b^2m}{n} \right)$.
- (7) If $^-$ is the angle between the tangents from $P(x_1, y_1)$ to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then
$$\tan^- = \frac{2ab\sqrt{\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1}}{x_1^2 + y_1^2 - (a^2 + b^2)}.$$
- (8) The tangents at the ends of the focal chord meet on the directrix.

• **Equation of Normal to Ellipse:**

- (1) **Point Form:** The equation of the normal at $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$.

- (2) **Slope Form:** If the line $y = mx + c$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $a^2m^2 + b^2 = c^2$. Hence the straight line $y = mx \pm \frac{m(a^2 - b^2)}{\sqrt{a^2m^2 + b^2}}$ always represents the normal to the ellipse.
- (3) **Parametric Form:** The equation of normal at any point $(a \cos \theta, b \sin \theta)$ is $ax \sec \theta - by \csc \theta = a^2 - b^2$.
- (4) The condition that the line $lx + my + n = 0$ to be normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{a^2}{l^2} + \frac{b^2}{m^2} = \frac{(a^2 - b^2)^2}{n^2}$.
- (5) Four normals can be drawn from any point to the ellipse and sum of the eccentric angles of their feet is an odd multiple of π .
- **Director Circle:** The locus of the point of intersection of the perpendicular tangents of an ellipse is a circle concentric with the ellipse. It is known as Director Circle.
 - The equation of the director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 + b^2$.
 - For the ellipse $\frac{(x - \alpha)^2}{a^2} + \frac{(y - \beta)^2}{b^2} = 1$, equation of the director circle is $(x - \alpha)^2 + (y - \beta)^2 = a^2 + b^2$.
 - **Auxillary Circle:** The locus of the feet of the perpendicular drawn from the foci to any tangent of the ellipse is a circle concentric with the ellipse. It is known as Auxillary Circle.
The equation of the Auxillary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2$ ($a > b$) and $x^2 + y^2 = b^2$ ($b > a$).
 - **Chord of Contact:** The equation of chord of contact of the point $P(x_1, y_1)$ with respect to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = 0$.
 - **Equation of Chord joining two points on an Ellipse:**
 - (1) The equation of chord joining the points with eccentric angles θ and ϕ on the ellipse is: $\frac{x}{a} \cos \frac{\theta + \phi}{2} + \frac{y}{b} \sin \frac{\theta + \phi}{2} = \cos \frac{\theta - \phi}{2}$.
 - (2) The equation of chord of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ having (x_1, y_1) as its midpoint is $\left(\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 \right) = \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 \right)$.

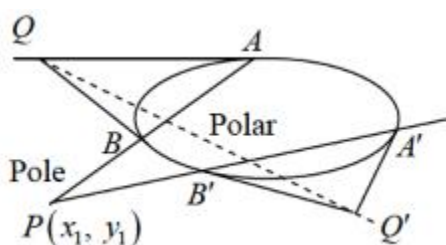
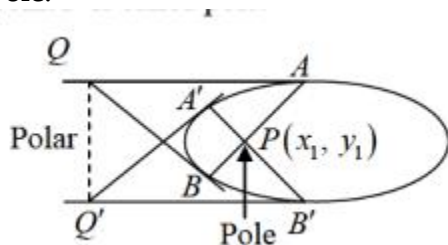
- (3) The point of intersection of tangent draw at points $P(\sim), Q(\)$ to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \left(a \frac{\cos\left(\frac{\sim +}{2}\right)}{\cos\left(\frac{\sim -}{2}\right)}, b \frac{\sin\left(\frac{\sim +}{2}\right)}{\cos\left(\frac{\sim -}{2}\right)} \right).$$

- (4) The point of intersection of normal draw at points $P(\sim), Q(\)$ to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \left(\frac{a^2 - b^2}{2} \cos \sim \cos \frac{\cos\left(\frac{\sim +}{2}\right)}{\cos\left(\frac{\sim -}{2}\right)}, -\frac{a^2 - b^2}{2} \sin \sim \sin \frac{\sin\left(\frac{\sim +}{2}\right)}{\cos\left(\frac{\sim -}{2}\right)} \right).$$

- **Equation of Pole and Polar:** Let $P(x_1, y_1)$ be any point inside or outside the ellipse. A chord through P intersects the ellipse at A and B. If tangents to the ellipse at A and B meet at Q, then locus of Q is called as **polar** of P with respect to the ellipse and point P is called as **Pole**.



- (1) The equation of polar of the point $P(x_1, y_1)$ with respect to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = 0.$$

- (2) The pole of the line $lx + my + n = 0$ with respect to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$\left(\frac{-a^2 l}{n}, \frac{-b^2 m}{n} \right), n \neq 0.$$

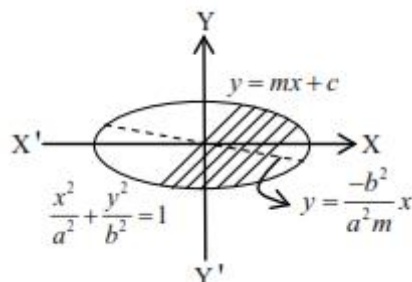
- (3) The pole of the line $lx + my + n = 0$ with respect to the ellipse

$$\frac{(x - \sim)^2}{a^2} + \frac{(y - \sim)^2}{b^2} = 1 \text{ is } \left(\sim - \frac{a^2 l}{l \sim + m + n}, -\frac{b^2 m}{l \sim + m + n} \right).$$

- **Properties of Pole and Polar:**

- (1) If the polar $P(x_1, y_1)$ passes through $Q(x_2, y_2)$, then the polar of $Q(x_2, y_2)$ passes through $P(x_1, y_1)$ and such points are called as **conjugate points**.
- (2) If the pole of a line $l_1 x + m_1 y + n_1 = 0$ lies on the another line $l_2 x + m_2 y + n_2 = 0$, then the pole of line $l_2 x + m_2 y + n_2 = 0$ lies on line $l_1 x + m_1 y + n_1 = 0$. Such lines are called as **conjugate lines**.

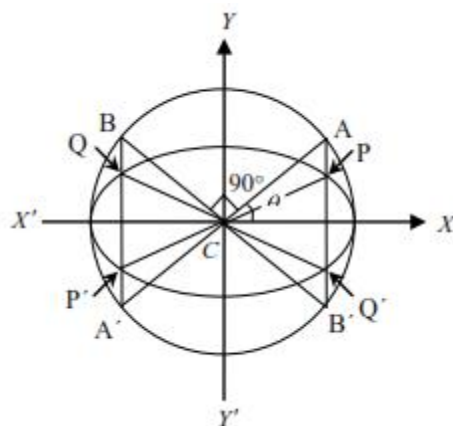
- (3) Pole of a given line is same as the point of intersection of tangents at its extremities.
- (4) The condition for two points (x_1, y_1) and (x_2, y_2) to be conjugate with respect to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{x_1 x_2}{a^2} + \frac{y_1 y_2}{b^2} - 1 = 0$.
- (5) The condition that two lines $l_1 x + m_1 y + n_1 = 0$ and $l_2 x + m_2 y + n_2 = 0$ to be conjugate with respect to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $a^2 l_1 l_2 + b^2 m_1 m_2 = n_1 n_2$.
- (6) The pole of the directrix of an ellipse is the corresponding focus.
- (7) The double ordinate which is conjugate to directrix of an ellipse is its corresponding latus rectum.
- **Equation of Pair of Tangents:** The equation of pair of tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\left(\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1\right)^2 = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right)\left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right)$.
 - **Eccentric Angle:** Let P be a point on an ellipse with centre C and NP be the ordinate of P. Let NP meet the auxiliary circle in P^1 . If $\angle NCP^1 = \theta$, then θ is called as eccentric angle of P. The point P^1 is called as corresponding point of P.
 - The point $P(a \cos \theta, b \sin \theta)$ traces the ellipse for all $0 \leq \theta \leq 2\pi$. P is denoted by the parameter θ . Hence parametric equation of ellipse are $x = a \cos \theta, y = b \sin \theta$. The corresponding point on auxiliary circle is $P^1(a \cos \theta, a \sin \theta)$.
 - If P_1 and P_2 are the ends of focal chord of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then $e \cos \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 - \theta_2}{2}$.
 - **Diameter of an Ellipse:** A line through the centre of an ellipse is called as a diameter of the ellipse.



- **Equation of Diameter of Ellipse:** The equation of diameter bisecting the chord $y = mx + c$ of slope m of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $y = -\frac{b^2}{a^2 m} x$ which passes through $(0,0)$.

• **Conjugate Diameter:**

- (1) Two diameters of an ellipse are said to be conjugate diameter if each bisect all parallel to the other. The co-ordinates of four extremities of two conjugate diameters are $P_1(a \cos \phi, b \sin \phi)$, $P_2(-a \cos \phi, -b \sin \phi)$, $Q_1(-a \sin \phi, b \cos \phi)$, $Q_2(a \sin \phi, -b \cos \phi)$.
- (2) If $y = m_1 x$ and $y = m_2 x$ be two conjugate diameters of an ellipse, then $m_1 m_2 = -\frac{b^2}{a^2}$.

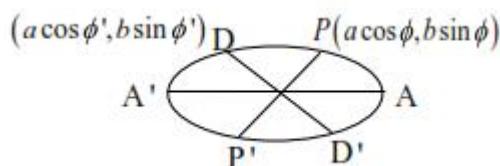


• **Properties of Diameters:**

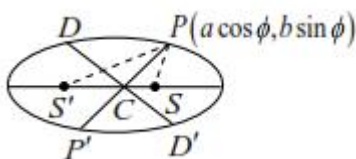
- (1) The tangents at extremity of any diameter are parallel to the chord it bisects or parallel to the conjugate diameter.
- (2) The tangents at the ends of any chord meet on the diameter which bisects the chord.

• **Properties of Conjugate Diameters:**

- (1) The eccentric angles of the ends of pair of conjugate diameters of an ellipse differ by a right angles i.e. $\phi' - \phi = \frac{\pi}{2}$.



- (2) The sum of the squares of any two conjugate semi diameters of an ellipse is constant and equal to the sum of the square of the semi axes of the ellipse. i.e. $CP^2 + CD^2 = a^2 + b^2$.

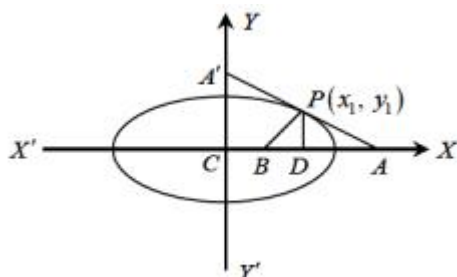


- (3) The product of the focal distances of a point on an ellipse is equal to the square of the semi diameter which is conjugate to the diameter through the point.
- (4) The tangent at the extremities of a pair of conjugate diameters form a parallelogram whose area is constant and equal to the product of the axes. i.e.
Area of parallelogram = $2a \times 2b$ = Area of rectangle contained major and minor axes.

- **Equi-Conjugate Diameter:** Two conjugate diameters are called as equi-conjugate, if their lengths are equal i.e. $CP^2 = CD^2$

$$\Rightarrow CP = CD = \sqrt{\frac{a^2 + b^2}{2}}$$

- **Length of Sub tangent & Sub Normal:** Let the tangent and normal at $P(x_1, y_1)$ meet the x-axis at A and B respectively.



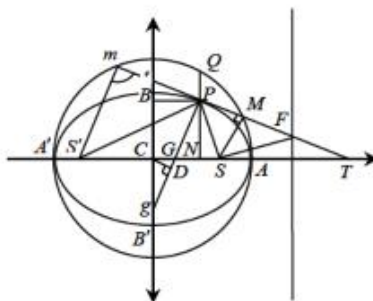
- (1) Length of sub tangent at $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$DA = CA - CD = \frac{a^2}{x_1} - x_1.$$

- (2) Length of sub normal at $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$BD = CD - CB = x_1 - \left(x_1 - \frac{b^2}{a^2} x_1 \right) = \frac{b^2}{a^2} x_1 = (1 - e^2)x_1.$$

- **Properties of Ellipse:**



- (1) $\frac{PN^2}{AN.A'N} = \frac{b^2}{a^2}.$
 - (a) $AS.A'S = b^2.$
 - (b) $CN.CT = a^2.$
 - (c)
- (2) The locus of the feet of the perpendicular M and m from the foci on any tangent to ellipse is the auxiliary circle.
- (3) The product of two perpendicular distances from the foci on any tangent of an ellipse is equal to b^2 i.e. $SM.S'm = b^2.$
- (4) $CG = e^2.CN = e^2x_1.$
- (5) The tangent and normal at a point on an ellipse bisect the angle between the focal radii to the point $SG = e.SP$ and $S'G = e.S'P.$
- (6) If the tangent on an ellipse meets the directrix in F, then PF subtends a right angle at the corresponding focus i.e. $\angle PSF = 90^\circ.$
- (7) Tangents at the ends of a focal chord intersect on the directrix.
- (8) If the normal at any point P meets the major and minor axes in G and g and CD is the perpendicular upon the normal, the $PD \times PG = b^2$ and $PD \times Pg = a^2.$
- (9) Tangents at the ends of any chord meet on the diameter which bisects the chord.
- (10) The sum of distances of any point P on the ellipse from the focus S and S' is $2a$ i.e. $PS + PS' = 2a.$
- (11) The ratio of y-co-ordinates of corresponding points on ellipse and Auxiliary circle is $b : a.$
- (12) The Harmonic mean of focal radii of any focal chord is equal to semi latus rectum $\frac{b^2}{a^2}.$
- (13) If $\sim, \text{¥}, \text{¢}$ be the eccentric angles of the four concyclic points on ellipse then $\sim + \text{¥} + \text{¢} = 2n\text{®}, n \in I.$
- (14) If $\sim, \text{¥}, \text{¢}$ be the eccentric angles of the feet P, Q, R and S of these normals then $\sim + \text{¥} + \text{¢} = (2n+1)\text{®}, n \in I.$
- (15) The necessary and sufficient condition for the normals at three points $\sim, \text{¥}$ on the ellipse to be concurrent if $\sin(\sim + \text{¥}) + \sin(\text{¥} + \sim) + \sin(\sim + \text{¥}) = 0.$
- (16) The maximum area of a rectangle that can be inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $2ab$ sq. units and the sides are $\sqrt{2}a$ and $\sqrt{2}b.$
- (17) Area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\text{®}ab$ sq. units.
